Response to Editor and Reviewer Comments

Manuscript TTE-Reg-2026-09-3162: "Machine learning in battery management systems: a structured scoping review and classification of 103 peer-reviewed studies, 2019–2026"

Reworked manuscript: "Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit"

We thank Professor Gao, the Associate Editors, and all four reviewers for a serious and, in the main, accurate set of critiques. The rejection identified real weaknesses: the corpus was curated rather than sampled while parts of the text read it as a field measurement; the validation-protocol taxonomy was under-defined and its arithmetic unreconciled; the classification rested on a single coder without a reliability statistic; the transportation-electrification fit was asserted rather than demonstrated; and the manuscript was not positioned against its closest prior review. Since receiving the decision we have substantially reworked the manuscript around those charges. This dossier responds point by point and is written to accompany a resubmission. Because the original submission was rejected rather than returned for revision, we state up front that the reworked manuscript differs from the version the reviewers saw in more than wording; the differences are listed in the next section and each is traceable to a specific reviewer comment.

A. What changed relative to the submitted version (disclosed in full)

- Retitled and redesigned as a two-tier scoping review. Tier 1 is the full 799-record rule-based eligible corpus (every eligible record included; no function quotas, year caps, or citation thresholds at this tier), from which bibliometric shares are measured with released rule-based function tagging and explicit capped-retrieval caveats. Tier 2 is the 103-study classified subset, whose quota-and-citation-ranked construction history is now disclosed in full and which is never generalized to the field. (R1.1, R2.1, R3-b1)
- The validation-protocol taxonomy is now defined before use, with two explicit stringency definitions (protocol-only and broader), an absorbing rule for borderline cases, and a released per-paper protocol audit (protocol_audit.csv) so every count reconciles row by row. (R1.2, R3-b6)
- A blind 30-paper re-coding reliability study was added, with per-field Cohen’s κ and Krippendorff’s α, an itemized disagreement table, locked and hash-tracked coder sheets, and a label-noise sensitivity analysis whose Monte Carlo noise model is now stated explicitly in the text. (R1.4, R2.2, R3-b4)
- A transportation-electrification cut of the corpus was added (fleet-tag share, fleet-data versus laboratory counts, venue concentration with numbers), together with a requirement-level mapping of BMS functions onto UN GTR 20, GB 38031-2020/2025, ISO 26262:2018, ISO/PAS 8800:2024, and ISO 21448. (R1.3, AE-1)
- The LLM discussion was corrected against its cited evidence, including the estimator role of the HA-LLM field-data study and the stationary-storage scope of the incident-analysis study, with a prior-art caveat. (R3-b7)
- A dedicated positioning against prior reviews was added, including a detailed comparison with the 2026 performance-centric EV-powertrain review of Prasanna et al. (Discover Artificial Intelligence 6:141), which two reviewers asked for. (R3-b8, R4)
- The complete evidence base is deposited at Zenodo (DOI 10.5281/zenodo.23436202) with a full artifact list; the AI-tool disclosure is consolidated into a single statement naming each tool and its purpose. (R1.4, R2 AI concern)
- The condensed submission form is 18 pages (ieee_submission/tex/mlbms.pdf, 105 references); an extended form (ML-BMS-Review.docx/pdf, 40 pp., 119 references, 13 tables) carries the same numbers with fuller discussion. All counts regenerate from the deposit.

B. Associate Editor and Area Editor Comments

AE-1. "As a review paper, this work does not cover sufficient directions of research or domains of applications. It also lacks sufficient outlook and insights into future research topics."

Coverage is extended along two axes. Horizontally, the Tier-1 eligible corpus measures the screened literature across all eight management functions (union-share bibliometrics over 799 records: SOH 20%, SOC 17%, RUL 13%, thermal 12%, fault/safety 12%, charging 8%, balancing 5% of records), so the review now reports domain coverage as a measurement rather than an assertion. Vertically, the validation audit, the standards mapping, and the per-function critical assessments add the analytical depth the AE found missing. On outlook: a new six-direction forward agenda (duty-cycle-aware validation; federated fleet evidence; benchmark diversification; pack- and vehicle-level ML aligned with UN GTR 20/GB 38031; functionally safe learning against ISO 26262/ISO/PAS 8800/SOTIF; honest foundation-model integration) replaces the earlier generic future-work paragraph (extended Section 9.8; condensed Section IX-C), and the reporting checklist is framed as an instrument that makes the next literature cycle auditable by construction.

AE-2. "The review manuscript does not demonstrate the level of technical depth, critical synthesis, and comprehensive state-of-the-art assessment expected for publication in an IEEE journal."

The rework adds the quantitative layer the AE called for: defined validation-protocol classes with per-paper assignments; reconciled counts (42 of 88 primary papers under strong protocols; 17 on information-leaking splits); a method-family contrast tested with Fisher's exact test, odds ratios, and 95% confidence intervals under two stringency definitions; and a label-noise sensitivity analysis with an explicit Monte Carlo model. The synthesis is now protocol-centric rather than narrative: each function chapter closes with a critical assessment of what the evidence actually supports (e.g., Sections 4.3, 5.6), and the safety-standards mapping converts the gap analysis into requirement-level statements an automotive reader can check. We accept that the submitted version fell short on these dimensions; we believe the reworked manuscript meets the bar, and we leave that judgment to the editors.

C. Reviewer 1

R1.1. The corpus is a curated reading list, not a systematic sample; every distribution in Section III is partly written in by the quota rule; either replace the quota step with a rule-based criterion applied to all 799 eligible records, or restate the Section III numbers as descriptive of this curated corpus and remove the field-level readings.

We did both, in a structured way. The 799 eligible records now constitute Tier 1 of the design and are all included, subject only to venue screening; the quota step survives only inside the Tier-2 in-depth subset, where its history is disclosed in full (stratified draw of 107 balancing functions and years; citation-ranked, deterministic within strata, walked deeper for recent years; 6 landmark seeds; −19 exclusions with reasons; +10 targeted additions; 103 = 98 pool + 5 seeds). Field-level claims now come only from Tier-1 measurements: function shares from rule-based title tagging with released term lists (validated against the subset's full-text labels, 71% agreement), and year counts presented as capped-retrieval counts with an explicit statement that they do not support absolute-growth claims. Tier-2 distributions are labeled as corpus composition in every section, and the abstract and conclusion no longer draw field-level readings from them. Section 2.3 states the within-stratum selection rule (deterministic citation ranking) that the reviewer correctly noted was missing.

R1.2. The central finding is not reproducible from the numbers as given; the strong-protocol category is never defined; per-protocol counts cannot be combined to yield the stated share; 10 versus 7 cross-chemistry studies are unrelated in the text; 15 reviews versus Table's 12; an audited table and a test on the contrast would settle this.

All four sub-points are fixed. (i) The taxonomy is now defined before use: four validation-protocol classes (within-cell split; unsupervised k-fold; cross-cell holdout with cross-chemistry and cross-domain extensions; online/real-vehicle validation), aggregated into strong (the isolation protocols) versus weak (the leaking splits), with environment-only categories (experimental hardware, simulation, case study) reported separately and never mixed into strong-protocol percentages; two stringency definitions are stated and every count says which it uses. (ii) The arithmetic now reconciles exactly and is auditable: 24 cross-cell + 7 cross-chemistry + 3 cross-domain + 8 online/fleet = 42 strong of 88 primary papers (48%); 12 within-cell + 5 k-fold = 17 weak; 29 environment-only (20 hardware, 9 simulation/case); per-paper assignments are released (protocol_audit.csv and the valid field of papers.csv) and were re-derived independently for 25 primary papers by the blind re-coding study. The 10-versus-7 relation is now stated: the 10 transfer validators comprise 7 cross-chemistry and 3 cross-domain, and the 7 are the studies behind the "validated across chemistries" count. (iii) The review count is reconciled as 12 method-family reviews plus 3 review-type syntheses coded under their dominant method family = 15 review-labelled papers, with the reconciliation carried in the terminology section and the function-by-year table's Reviews column. (iv) The contrast is tested: under the protocol-only definition, physics-informed 8/22 (36%) versus others 34/66 (52%), Fisher exact p = 0.32, OR 0.54 (95% CI 0.21–1.46); under the broader definition, 13/22 (59%) versus 49/66 (74%), p = 0.19, OR 0.50 (95% CI 0.19–1.36); neither is statistically significant, both are reported, and the text claims neither as a model-family effect.

R1.3. The scope is not yet transportation electrification; the standards appear as motivation and never return; a mapping from BMS function to safety-standard requirement would belong here.

The rework adds exactly that mapping, as a table and as an inference rule. Each BMS function is mapped to what UN GTR 20 (including the Amendment 1 thermal-propagation provisions), GB 38031-2020/2025, ISO 26262:2018 with ISO/PAS 8800:2024, and ISO 21448 demand of a deployed function, alongside the corresponding subset status (e.g., 3 of 13 SOC primary papers under a strong isolation protocol; 0 of 8 charging and 0 of 7 thermal primary papers; 2 of 7 fault/safety papers online on fleets). The section states its inference rule explicitly: within-cell and k-fold evidence cannot support a standards argument because the safety case is about unseen cells and unseen duty. The corpus itself is now cut for this readership: about 2% of eligible records carry fleet/cloud tags; 10 of 88 primary papers use real vehicle or fleet data against 36 on laboratory cells; venue concentration is quantified (Journal of Energy Storage 103 of 799 eligible records; IEEE-prefixed venues 9.0%); and the forward agenda names pack- and vehicle-level ML as a first-class direction.

R1.4. DOI resolution verifies existence, not metadata; reference [10] is attributed to an RFID journal; the artifact link is a GitHub account rather than a DOI-backed deposit; classification was one coder with a corrective second pass; a blind double-coding with an agreement statistic would cost little.

Each item is addressed. (i) The verification claim is now scoped to what DOI resolution does, in the limitations text. (ii) The RFID-venue record is flagged in the manuscript itself rather than silently dropped, and the released audit file (final_audit.json) carries each reference's Crossref-resolved title and venue at retrieval time so such artifacts are visible rather than hidden. (iii) The artifact of record is now a Zenodo deposit, DOI 10.5281/zenodo.23436202, containing the classified dataset, the eligible pool, the record-level screening log, the protocol audit, the codebook, the reliability files, the conference set, the sensitivity script, and all generation scripts; the GitHub repository is described as a development mirror, which is what item 12 of our own checklist asks of others. (iv) The blind double-coding the reviewer proposed was run: two co-author coders not involved in the primary coding re-coded a 30-paper stratified random sample under the released codebook, blind to the released labels, to the population statistics, and to each other, with sheets locked and hash-tracked at completion. Agreement is strong on three of five fields (function $\kappa = 0.92$; method family $\kappa =$

0.88; data source κ = 0.88) and below threshold on chemistry (κ = 0.55) and validation protocol (κ = 0.48), with the evidence-access constraint disclosed (full-text access to 7 and 2 of the 30 papers). Because the coders are authors of the study, the manuscript states plainly that the blinding was procedural rather than external, and all headline validation counts carry a sensitivity analysis built from the observed coder-versus-label disagreement.

D. Reviewer 2

R2.1. Function quotas, year balancing, targeted additions, and quota-based exclusions shaped the final 103 papers; publication trends, application shares, and method-family growth cannot be interpreted as field-wide trends.

Agreed, and the rework enforces it structurally. Trends and shares are now either measured on the quota-free 799-record Tier-1 pool (function union shares; capped-retrieval year counts) or explicitly labeled as within-subset patterns. The abstract, Section 3, and the conclusion were rewritten so that no field-level reading rests on the curated subset; e.g., the method-family evolution paragraph states the primary-paper counts year by year with the partial-2026 caveat and calls the trajectory a within-subset observation, and the conclusion phrases the field-level statement only where Tier-1 measurement supports it.

R2.2. One author coded all papers; the co-author review was corrective rather than independent; without independent double coding and an agreement measure it is unclear how much of the pattern reflects coder decisions.

The reliability study now exists and is reported without embellishment: 30-paper stratified random sample (seed-pinned, reproducible), released codebook with a decision tree and an absorbing rule, two co-author coders blind to the released labels and to each other, per-field κ and α, an itemized disagreement table, and locked hash-tracked sheets. The manuscript states the limits honestly: the coders are authors (procedural, not external, blinding), and their full-text access was uneven, which the two low-κ fields (chemistry, validation protocol) plausibly reflect. Quantitatively, all headline validation counts now carry a label-noise sensitivity analysis: the strong-protocol share spans 40–59% under plausible label error, and no qualitative conclusion depends on where in that range the truth lies. The noise model is stated in the text (4,000 seeded Monte Carlo draws; per-draw flip rate ~ Normal, centered on the observed coder-versus-label bucket error rate of 0.35, SD 0.02, truncated at zero) and the script is released.

R2.3. The paper does not demonstrate that within-cell or k-fold validation overstates accuracy; it identifies a methodological risk, not direct evidence of inflated accuracy.

The reviewer’s reading is the one we now adopt verbatim in substance: the review claims a methodological risk, not measured inflation. The text was purged of causal accuracy claims: within-cell and k-fold splits are said to leak trajectory information and to produce errors that do not survive cell-to-cell variability — a statement about protocol design — and the conclusion phrases the finding as the proliferation of published accuracy claims having outpaced the validation evidence behind them, not as a demonstration that reported accuracies are inflated. We also explain, in Section 5.6, why cross-paper pooling of reported errors is not defensible for this literature (metric heterogeneity), which is precisely why a between-group accuracy comparison was not attempted.

R2.4. The physics-informed claim (13/24 versus 49/67) has no confidence interval, significance test, or adjustment; it may reflect corpus composition; it should be treated as exploratory.

Fixed as specified. Both stringency definitions are reported with Fisher’s exact test, odds ratios, and 95% confidence intervals (protocol-only: 8/22 = 36% versus 34/66 = 52%, p = 0.32, OR 0.54, CI 0.21–1.46;

broader: 13/22 = 59% versus 49/66 = 74%, p = 0.19, OR 0.50, CI 0.19–1.36); the text states that neither difference is significant, that confounding is unadjusted (physics-informed work skews early-stage and simulation-heavy), that n = 22 is small, and that the result is an untested hypothesis rather than a settled finding — the same words appear in the conclusion. The label-noise sensitivity analysis leaves the contrast non-significant in 95% of draws.

R2.5. The conference sensitivity analysis compares abstract-level evidence with fuller journal assessment; a missing validation statement in an abstract does not mean validation was absent; the set is small and citation-ranked.

The conference section is reframed as what it can be: a descriptive check of stated-evidence transparency, not a test of venue-selection bias. The comparison is now between abstract-stated validation on the conference path and abstract-stated experimental-hardware validation among journal papers' indexed records, with the truncation caveat the reviewer raised stated in the text, the pre-stated mechanical inclusion rule applied to all 14 retrieved papers (the conventional model-based estimators the reviewer noted are excluded from the learning-path comparison by that rule and retained for transparency), and the set released in full. The section's conclusion is now about reporting transparency, not about conference practice, and it feeds the checklist rather than any venue claim.

E. Reviewer 3

R3-b1. Function-quota selection introduces selection bias; proportions cannot establish which functions or methods dominate the literature; claims should be restricted to the selected corpus.

Restriction is now structural: the Tier-2 subset is never generalized, and where dominance claims are made they are Tier-1 measurements (rule-based tagging over all 799 eligible records, term lists released, tagger validated at 71% against the subset's full-text labels). The two levels are labeled everywhere they appear, including figure captions.

R3-b2. The selection process is difficult to examine; 107 candidates, 19 exclusions, and 15 additions reach 103 but do not explain which studies were removed for ineligibility versus balance; a record-level screening table with reasons would make selection verifiable.

The record-level table exists and is released: screening_log.csv carries all 1,550 screening decisions with outcomes and reasons, the selection files identify every inclusion and drop at every stage with the five forced seeds annotated, and Section 2.3 of the manuscript now narrates the funnel with the categories of exclusion (corrigenda, misindexed records, scope mismatches, quota-driven trimming) distinguishable. The flow figure matches the narration, including the 98 + 5 decomposition.

R3-b3. Section II-B admits reviews only to the cross-cutting cluster, yet Table II assigns reviews to several categories; Table I reports 12 reviews while Section III-C mentions 15.

The definitions are reconciled and stated once, then used consistently: 12 method-family reviews are admitted under the review cap, and 3 further review-type syntheses are coded under their dominant method family, giving 15 review-labelled papers; the function-by-year table carries a Reviews column that states this reconciliation per function, and the terminology section defines primary-application (88) versus review-labelled (15) papers. Where the earlier text said reviews sit only in the cross-cutting cluster, it now says reviews are admitted to the cross-cutting cluster and that review-type syntheses are coded under their dominant family.

R3-b4. The coding process lacks an independent reliability check; a second independent coding of a subset with a codebook and dispute rules would establish reproducibility.

Done as proposed, with the honesty limits stated: 30-paper stratified random sample, released codebook with decision tree and absorbing rule, two co-author coders blind to the released labels and to each other, locked hash-tracked sheets, per-field κ/α, and an itemized disagreement table (see the response to R2.2). The manuscript does not claim external independence it does not have; it quantifies what procedural blinding yields and attaches sensitivity ranges to every headline count the coding supports.

R3-b5. Method-evolution analysis is weakened by uneven sampling across years (one 2019 landmark, seven 2026 papers, searches ending early September); Figure 3 displays percentages despite small, differently selected denominators; displacement claims should be expressed as within-sample patterns.

The evolution figure now displays paper counts, not percentages; the 2019 seed is shown as what it is (a forced landmark); 2026 is labeled a partial year at the September retrieval date; and the physics-informed trajectory is stated as primary-paper counts (1, 2, 3, 3, 4, 4, 5 across 2020–2026) with the monotonicity remark explicitly confined to the within-subset primary trajectory. The measured-tier year counts (Tier 1) are reported separately, with the 2024 retrieval hole documented, and the text states that neither level supports absolute-growth claims.

R3-b6. Cross-cell holdout, cross-domain transfer, laboratory hardware, and fleet operation test different properties; coding rules for borderline cases are unclear; the classification underlying the central comparison should specify thresholds, study-level assignments, and distinguish generalization from deployment evidence.

The codebook now specifies each threshold and the absorbing rule (a study is credited with the strongest protocol it reports for its headline claim), with a decision tree and worked examples; study-level assignments are released (protocol_audit.csv); and generalization evidence (isolation protocols) is distinguished from deployment evidence (operating environment) by construction — the environment-only categories are reported separately and never mixed into strong-protocol percentages. The fleet table additionally separates data use (10 papers on real vehicle/fleet data) from online validation (8 papers), which are different measures, and cross-tabulates them.

R3-b7. The LLM discussion does not match its evidence: reference [15] is an LLM-based SOH estimator on field data with uncertainty-aware controls, not merely an orchestration layer; reference [14] concerns stationary-storage incidents, not onboard EV control; Table V should distinguish what the studies demonstrate from the authors' proposed restrictions.

Correct on both counts, and the section is rewritten accordingly. The HA-LLM study is now described as what it is: an LLM-based SOH estimator that aligns resampled charging-current sequences with the LLM embedding space, injects degradation priors by prefix adaptation, and wraps the model in uncertainty-aware Monte Carlo gating with substitution by a physics-consistent estimator, validated on one year of field data from 100 production vehicles with cross-vehicle few-shot and zero-shot evaluation — i.e., it already occupies the estimator seat, under exactly two of the guardrails we propose. The incident-analysis study is described as stationary, station-side technology, not onboard control. A prior-art caveat was added (LLM-based battery estimation exists in preprints, conferences, and a 2025 journal letter outside the screened venues), and Table 9 now states explicitly which column reports what the studies demonstrate and which column is our normative guidance.

R3-b8. A 2026 systematic review has already classified 95 studies across battery-management applications and computational methods; earlier reviews examined similar approaches; the contributions are not clearly distinguished; a more direct comparison is needed.

A positioning section now does this directly (extended Sections 1.3 and 9.7 with a comparison table; condensed Sections I-B and IX-C with Table VII). The closest recent cross-functional review — the

performance-centric review of Prasanna et al., Discover Artificial Intelligence 6:141 (2026), covering 240 publications across the EV powertrain — is compared in detail: in scope (BMS functions in depth versus powertrain breadth), in unit of analysis (validation-protocol audit versus aggregation of reported RMSE/R^2), and in question asked (which claims can be checked, versus which algorithm family performs best as reported). The comparison is framed as complementary rather than adversarial: a performance-centric aggregation inherits, unmeasured, exactly the validation weaknesses this review quantifies. The same section positions the work against the function-specific and method-specific reviews cited in the introduction, and the contribution list now states the delta explicitly.

F. Reviewer 4

R4. "A thorough review, however, it remains incremental in nature. Numerous reviews are available, to cite one: Prasanna et al. (2026), Discover Artificial Intelligence, 6(1), 141."

The detailed response to R3-b8 applies, and the comparison with Prasanna et al. is now in the manuscript in the reviewers' requested form: a dedicated paragraph plus a per-axis table. In one sentence, the delta is: prior reviews, including the performance-centric powertrain review, synthesize what is claimed and how well it is reported to perform; this review additionally measures how those claims are validated (protocol classes, per-paper assignments, method-family contrast with significance tests and sensitivity ranges), ties the gaps to three automotive safety standards at requirement level, and releases the record-level dataset, screening provenance, codebook, and re-runnable pipeline that make every count auditable. We believe that combination — not any single finding — is the contribution, and it is the one no available review offers.

G. Reviewer 2's assessment that the manuscript gave "the strong impression that it was generated almost entirely with AI"

We take this concern seriously and answer it with disclosure and evidence rather than assurance. The manuscript now carries a single consolidated declaration naming each tool and its purpose: an AI language model (GLM, Z.ai) assisted in drafting and adapting the literature-pooling, analysis, and figure-generation scripts; ChatGPT, Grammarly, and QuillBot were used for language editing. The declaration also states what no AI tool did: no AI tool decided study inclusion, and no AI tool assigned or altered any classification label — all screening decisions and all released labels were made by the authors, and the released reliability files (locked coder sheets, blinding memo, disagreement tables) document the human coding evidence. Every reference was verified by direct DOI resolution, every count and figure regenerates from the deposited dataset, and the authors take full responsibility for the content. We hope the reworked methodological substance — defined taxonomies, tested contrasts, disclosed limitations — also addresses the underlying judgment behind the impression.

H. Summary

The rejection identified that the manuscript asked the reader to trust numbers the design could not support. The rework fixes the design rather than the wording: a two-tier evidence base that separates measurement from curation, a validation taxonomy with audited per-paper assignments and tested contrasts, a reliability study with its limits stated, a transportation-electrification cut with a safety-standards mapping, a corrected LLM discussion, an explicit positioning against prior reviews including the one two reviewers named, and a DOI-backed deposit behind every number. We believe the reworked manuscript is a different and substantially stronger paper, and we would be glad to submit it to the Transactions as a new manuscript if the editors consider that appropriate; otherwise this dossier accompanies its submission elsewhere.

Sincerely,
The authors

Leading University, Sylhet, Bangladesh — 18 October 2026